Oil Cooler (Mod 6/1438 and Mod 6/1481) – Maintenance Practices

1. Removal/Installation

Refer to Figure 1 for details.

A. Remove Oil Cooler

1) Lower engine bottom cowling.

2) Have container in position to receive oil spillage.

3) Disconnect inlet and outlet lines at oil cooler. Cap open lines.

4) Remove bolts securing drain lines bracket to lower rear flange of oil cooler. Disconnect clips from drain lines and remove bracket.

5) Remove bolts securing oil cooler to duct assembly. Remove oil cooler by sliding cooler forward and down.

B. Install Oil Cooler

1) Position oil cooler to duct assembly, center on duct opening and install eight retaining bolts, washers and nuts.

2) Clip drain lines to drain lines bracket and position bracket on lower rear flange of oil cooler.

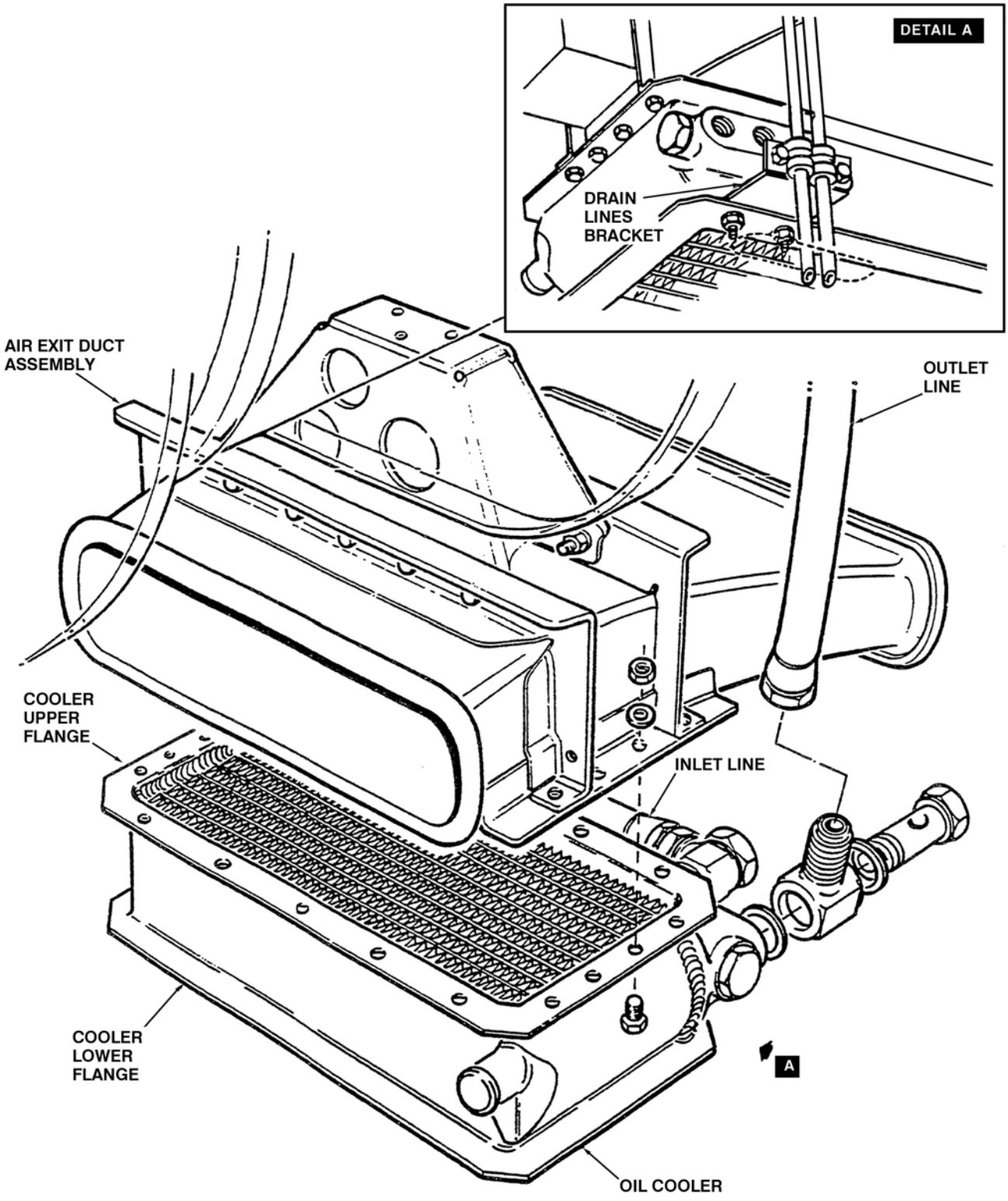
3) Raise engine bottom cowling and ensure drain lines protrude through holes in cowling. Lower cowling and install two bracket retaining screws, washers and nuts.

Note If installing new oil cooler, drill cooler upper and lower flanges to match holes in duct (Para 1.B.1)) and drain lines bracket (Para 1.B.3)).

4) Connect inlet and outlet lines to oil cooler.

5) Fill oil tank. (Refer to 12-10-25, Oil System – General Data).

6) Check for oil leaks on subsequent engine run. Top up oil tank after engine run.



Oil Cooler Installation
Figure 1

2. Adjustment/Test

A. Bench Test Oil Cooler

Equipment required:

- Liquid bath
- Regulated pressure air source
- Hot water supply
- Test setup as shown in Figure 2

1) Plug cooler outlet and attach air hose to inlet.

2) Attach other end of air hose to regulated pressure air source and pressurize cooler to 12 psi.

3) Submerge cooler in water at 140° – 180°F.

4) After five minutes of immersion, increase air pressure to 100 psi. There must be no evidence of leakage.

WARNING **INCREASE AIR PRESSURE GRADUALLY TO AVOID BURNS FROM HOT WATER SPLASHING, IF LARGE LEAKS OCCUR.**

5) Slowly vent air pressure, remove cooler from bath and disconnect air hose.

6) Install cooler in test setup as shown in Figure 2.

7) Apply room temperature water at 150 psi (max) to cooler and hold.

8) After retaining pressure for 10 minutes, check cooler for blown or bulged plates.

9) Slowly release pressure and remove cooler from test setup.

10) Dry cooler thoroughly with compressed air.

11) Test Mod 6/1481 control valve function as follows:

a) Submerge valve in water at 160 ± 5°F for five minutes.

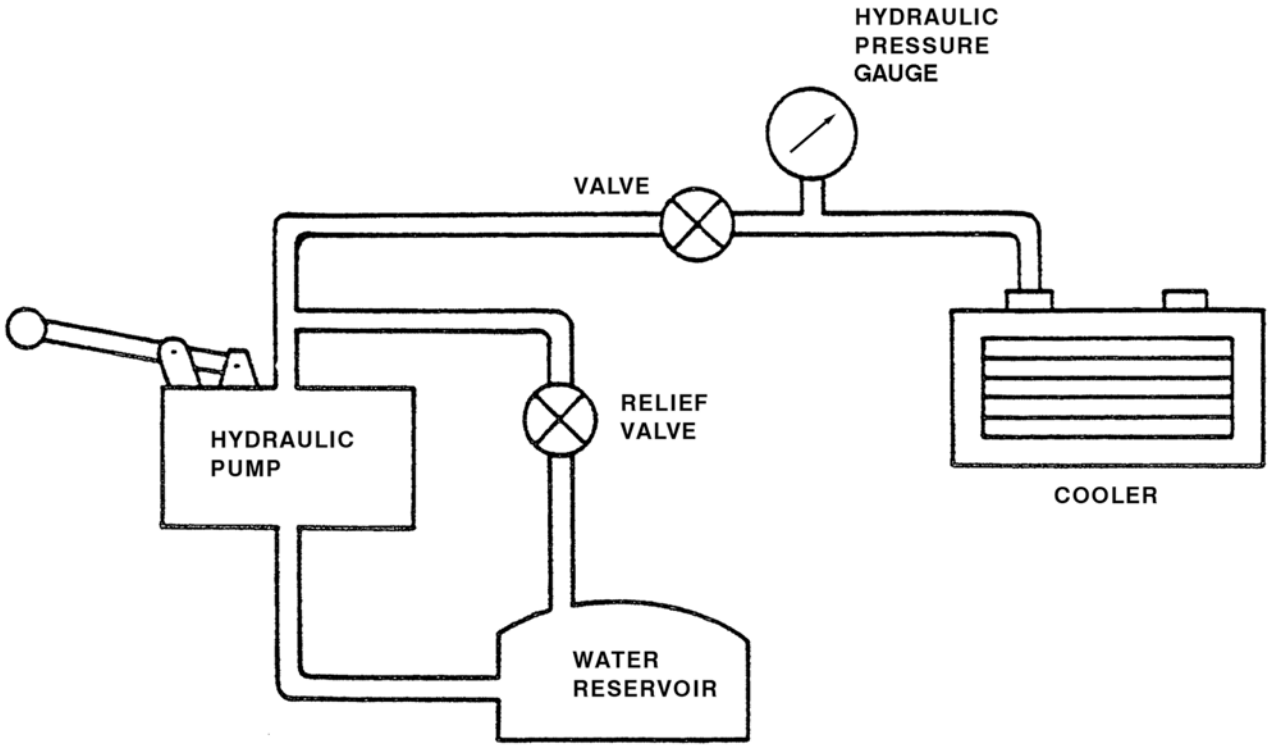
b) Remove valve, measure and record length.

c) Submerge valve in water at 185 ± 5°F for five minutes.

d) Remove valve, measure and record length.

e) Check difference in lengths recorded in Para 2.A.11).b) and Para 2.A.11).d) is not less than 0.170 inch.

12) Dry valve and cooler thoroughly with compressed air.



Hydrostatic Test Setup
Figure 2